Industry 4.0 and 5.0

Twenty Fifth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



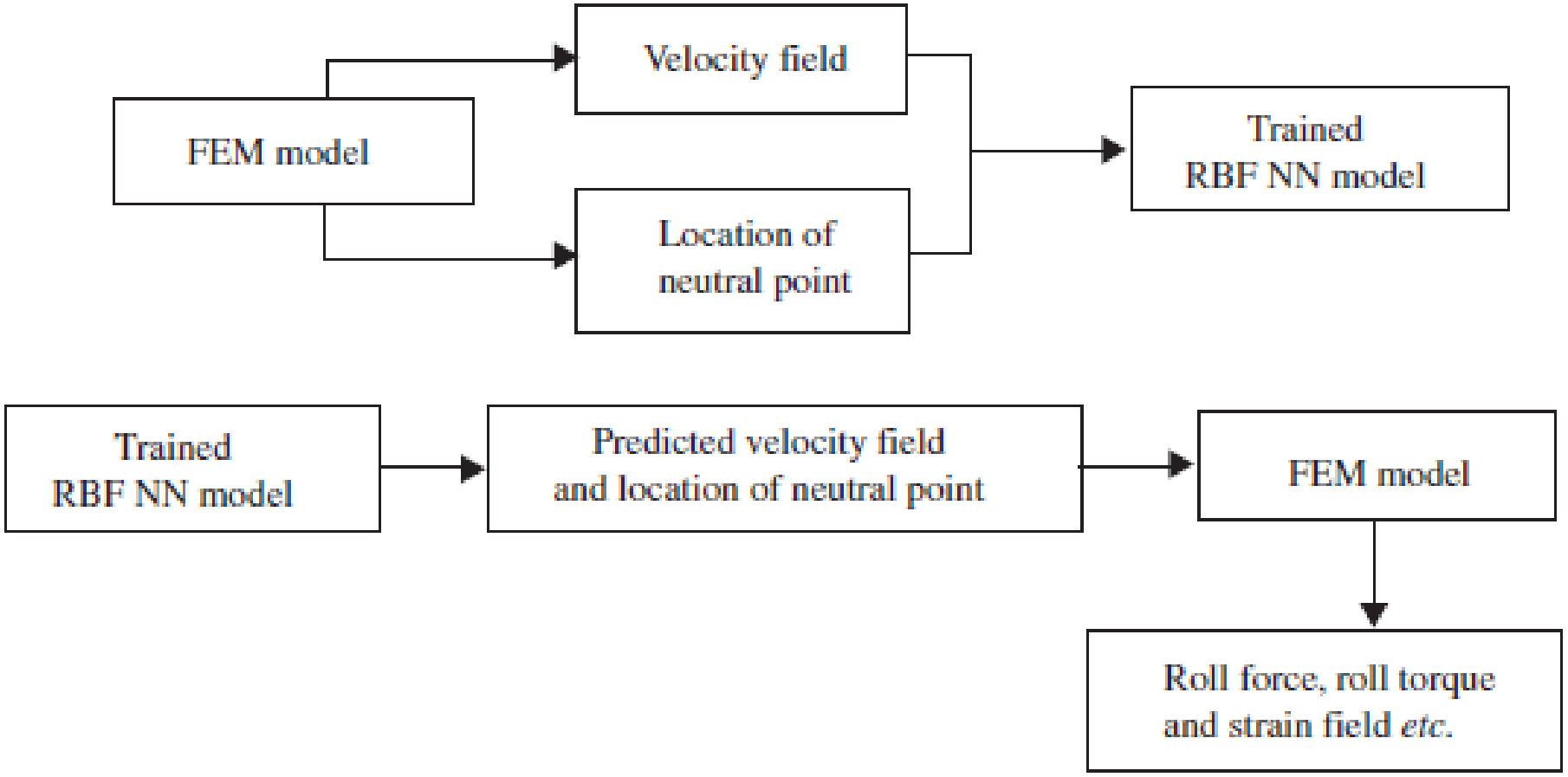
Industry 4.0





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Hybrid computational techniques have promise



FE simulation of FSW

- FE simulations of FSW have been performed using DEFORM-3D.
- DEFORM-3D capabilities:
  - Coupled temperature-displacement analysis can be carried out.
  - Stress, strain, strain rate, temperature distribution can be achieved.
  - Material flow and velocity of each point can also be visualised.
  - Thermal, mechanical and metallurgical responses of parts during exposer to heat can also be predicted. Thermal distortion, quench cracking and residual stresses can be predicted.
- However, **FE simulation of FSW is computationally very expensive.**

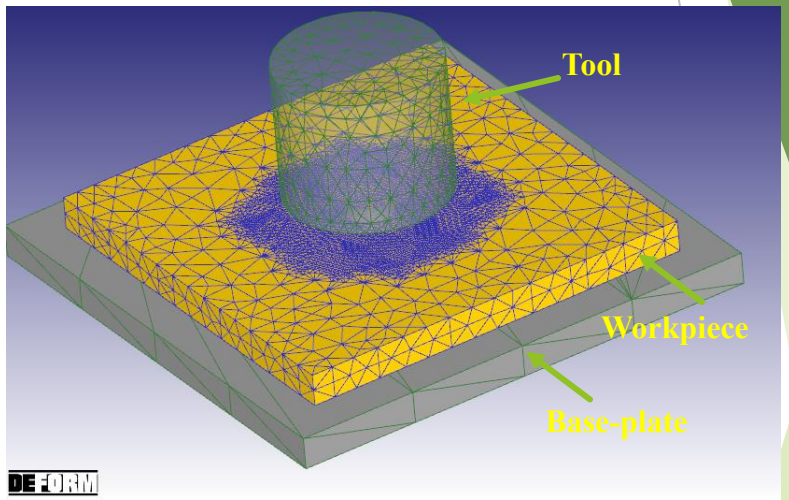


Fig: FE simulation of FSSW

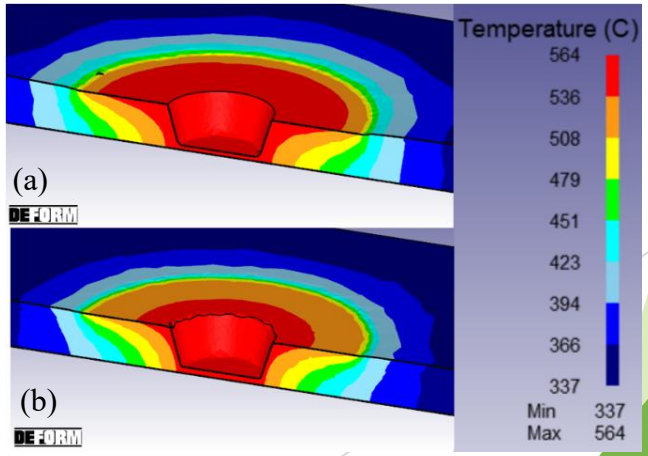


Fig: Temp distributions contours during FE simulation of FSSW (a) unlubricated (b) lubricated conditions

FE simulation time for FSW

- The modelling of FSW process is complex due to the numerous variable parameters applicable during the process
- Literature available on simulation time of FSW

Authors	Formulation	Simulation details	Time taken
Lasley (2005)	Arbitrary Lagrangian Eulerian (ALE)	5.44 second plunging using Forge3 in supercomputing facility	42 days
Awang (2007)	Lagrangian	1.5 seconds simulation time using 3.5 GHz intel Pentium 4	14 days 12 hours
Assidi et al. (2010)	Arbitrary Lagrangian Eulerian (ALE)	6.6 seconds simulation in Forge3 software package	12 days 20 hours
	Eulerian		6 days 14 hours
Mallik et al. (2014)	Coupled Eulerian Lagrangian (CEL)	15 seconds simulation using ABAQUS/Explicit	9.5 hours

- Simulation time has been reducing with improvement in computational facility available. Following are simulation time using DEFORM-3D:

Stages (experimental time)	Simulation details with time taken	
	Lagrangian/ Conjugate gradient iterative solver/Newton-Raphson iteration	Lagrangian/ Sparse solver/Direct iteration
Plunging (6.33 seconds)	5 hours	3 hours
Dwelling (5 seconds)	3 hours	2 hours
Traverse (37 seconds)	10 hours	9 hours
Total (48.33 seconds)	**18 hours**	**14 hours**

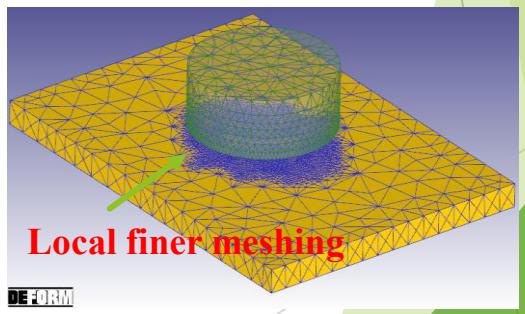


Fig: Finer local meshing to reduce simulation time

- **There is a requirement of study on reducing simulation time for practical utilization of FSW FE simulation.**

Meshless Methods

- **Meshless methods:**

  - Group of numerical methods for solving partial differential equations on regular or irregular distribution of points.
  - Costly mesh generation and remeshing is not required.
  - Some methods are based on weak form of differential equations, whilst others use collocation points.
  - These methods lack in robustness and computational speed, even though in some cases, the quality of solution has been shown to be better than that obtained by FEM .
  - These methods have not been extensively applied for solving the problems in metal forming industry.

Meshless Methods

- Reproducing kernel particle method (RKPM):

  Chen *et al.* have used RKPM for the analysis of ring compression and upsetting problem.

  Shangwu *et al.* utilized RKPM method for modeling of plane strain rolling problem.

- Corrected smooth particle hydrodynamics (CSPH):

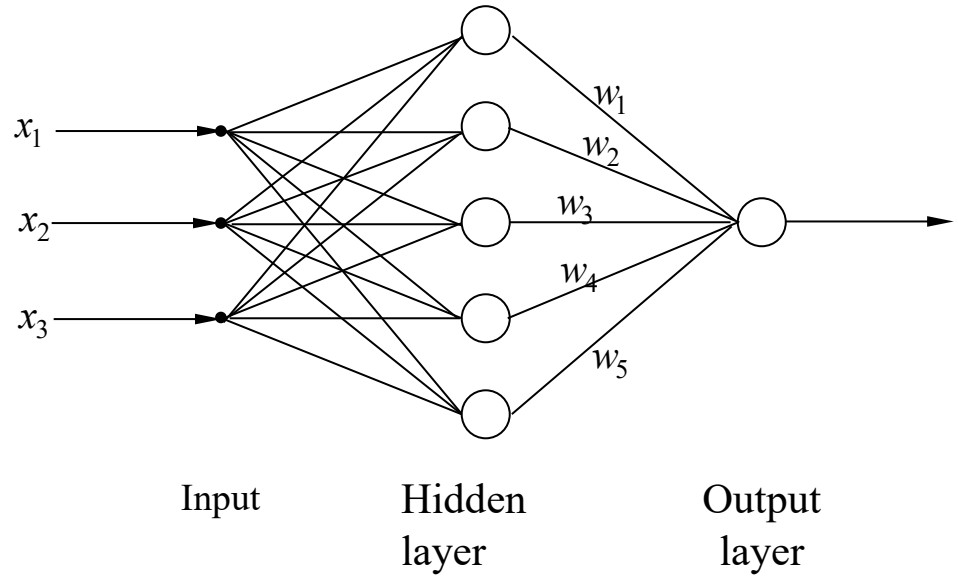
  Bonnet and Kulasegaram *et al.* utilized the CSPH to perform two-dimensional simulations of several metal forming processes without considering strain-hardening.

- Element free Galerkin method :

  Xiang *et al.* applied element free Galerkin method for simulation of plane strain rolling.

Meshless Methods

- **Radial basis function collocation method:**
  - An axisymmetric forging problem has been analyzed using this method.
  - In this method, Boundary conditions and governing equations are satisfied at the collocation points.



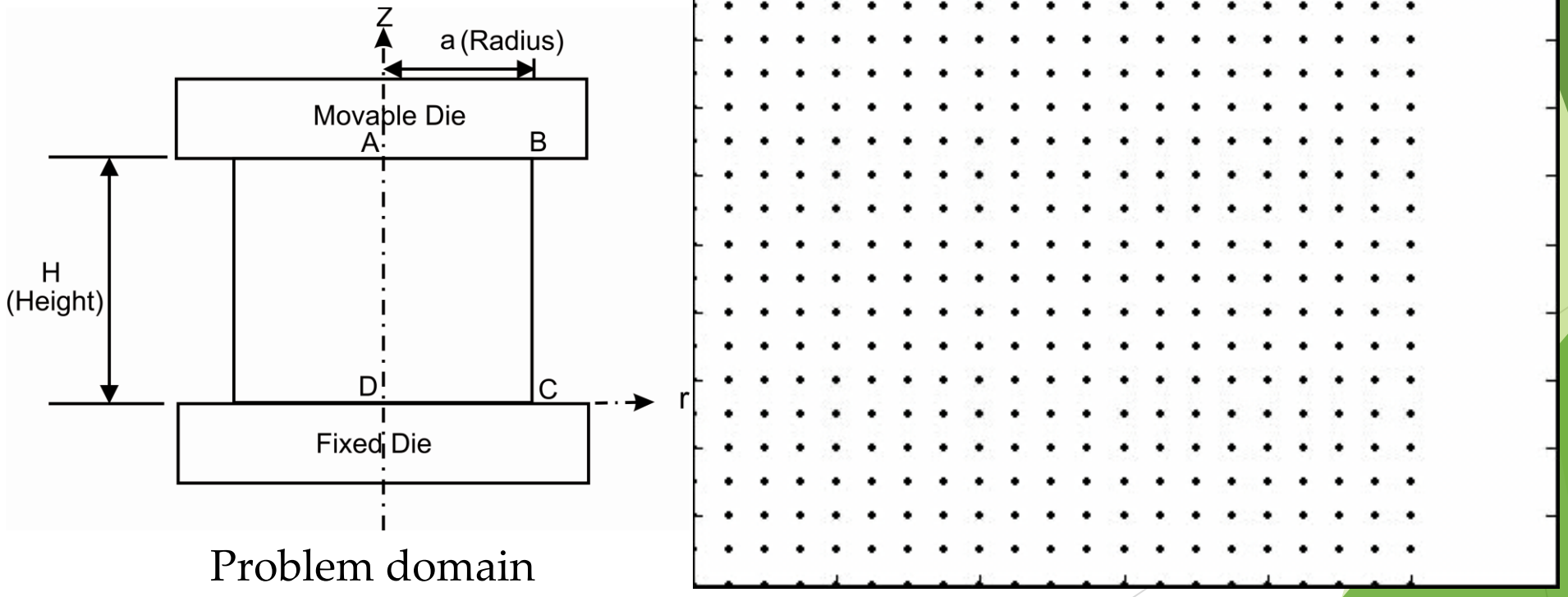
Architecture of a typical radial basis function neural network

$$\phi = \sum w_i g_i \left(\| \boldsymbol{x} - \boldsymbol{x}_k \|, c_k \right)$$

- Multiquadric $g = \sqrt{r^2 + {c_k}^2}$
- Gaussian $g = \exp\left(-\frac{r^2}{{c_k}^2} \right)$
- Inverse multiquadric $g = \frac{1}{\sqrt{r^2 + {c_k}^2}}$
- Thin-plate splines $g = r^{2m} \ln r$

Simulation of the process using RBF collocation method

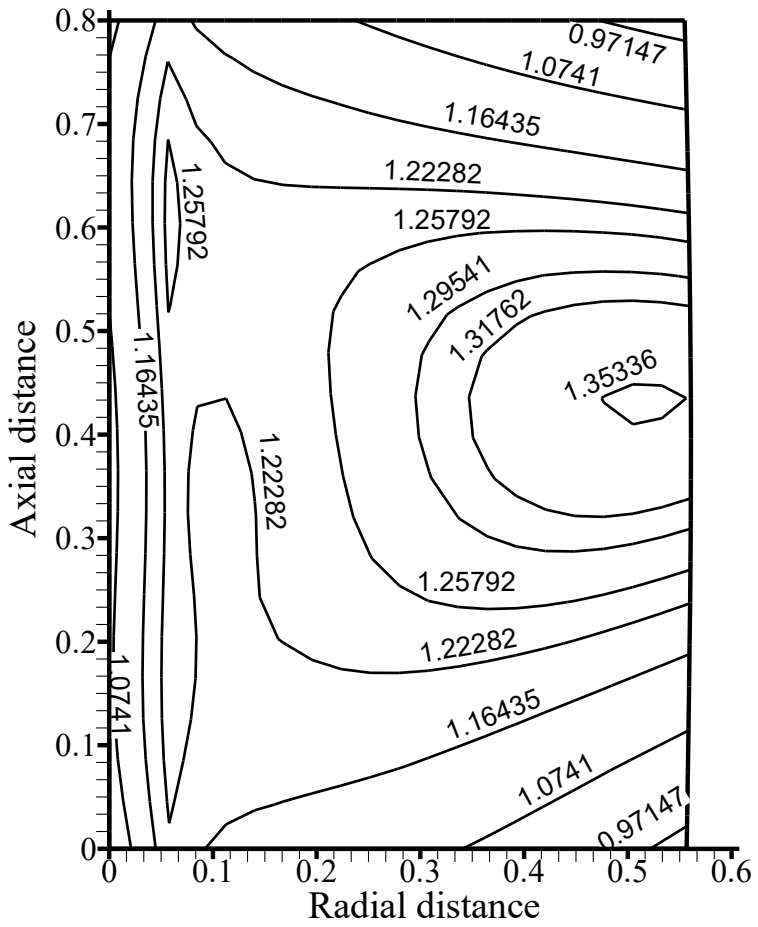
- A typical example of an axisymmetric forging problem:



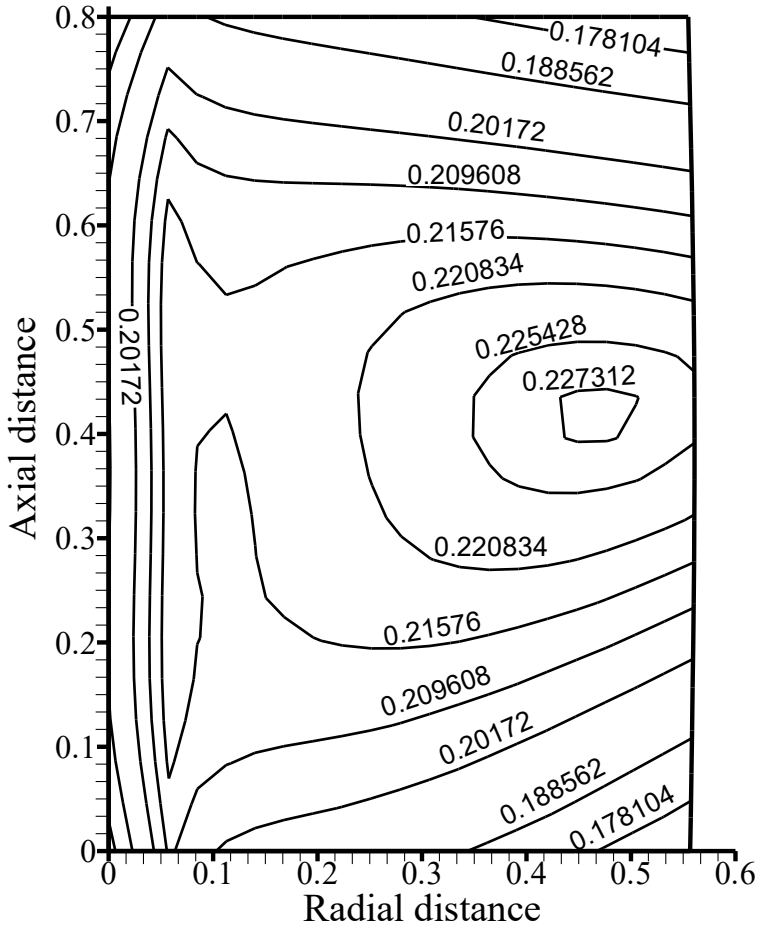
Problem domain

Int J Adv Manuf Technol (2007) 34:464–473

- **Typical contours:**



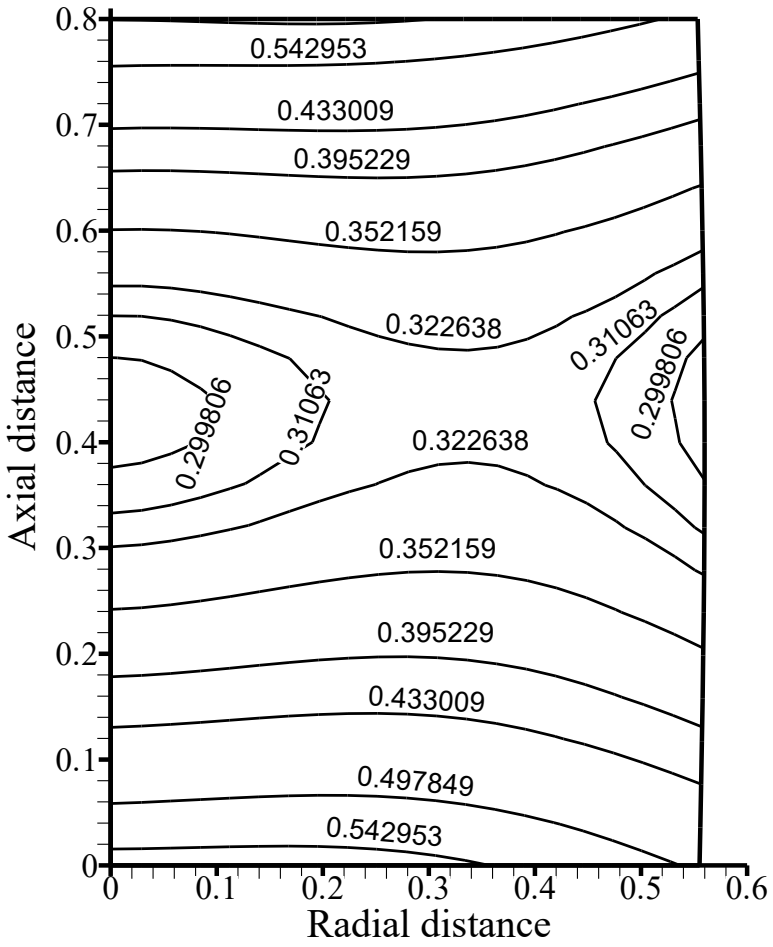
Non-dimensional equivalent strain rate at 20 % reduction in height with $f = 0.05$ and $R/H = 1/2$.



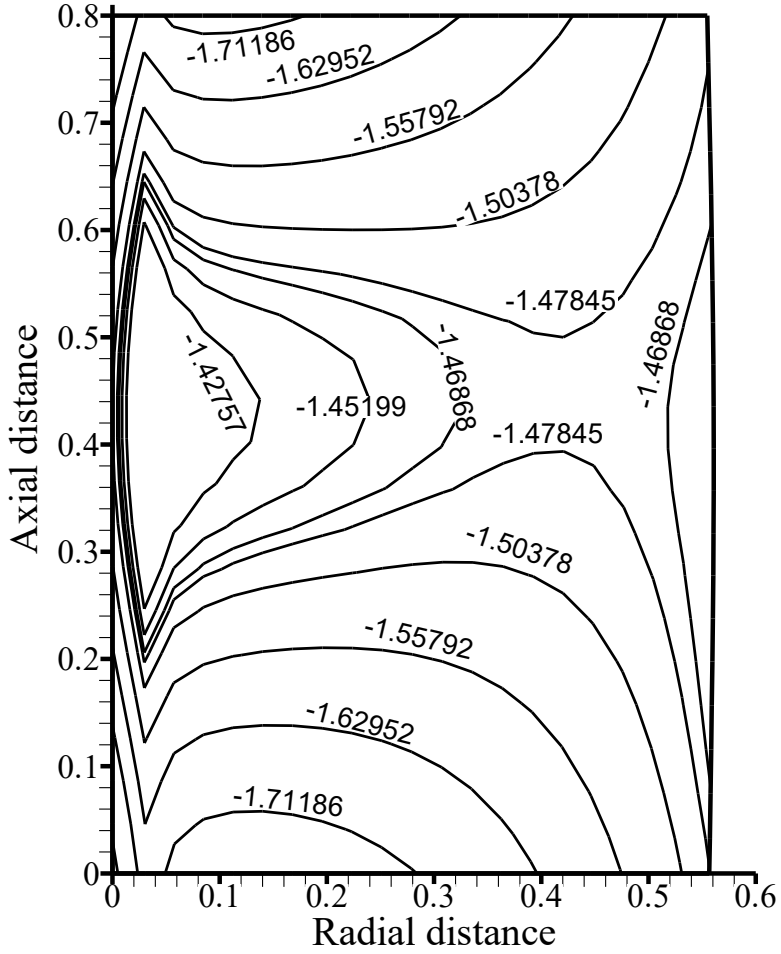
Equivalent plastic strain at 20 % reduction in height with $f = 0.05$ and $R/H = 1/2$.

Meshless Methods

Int J Adv Manuf Technol (2007) 34:464–473



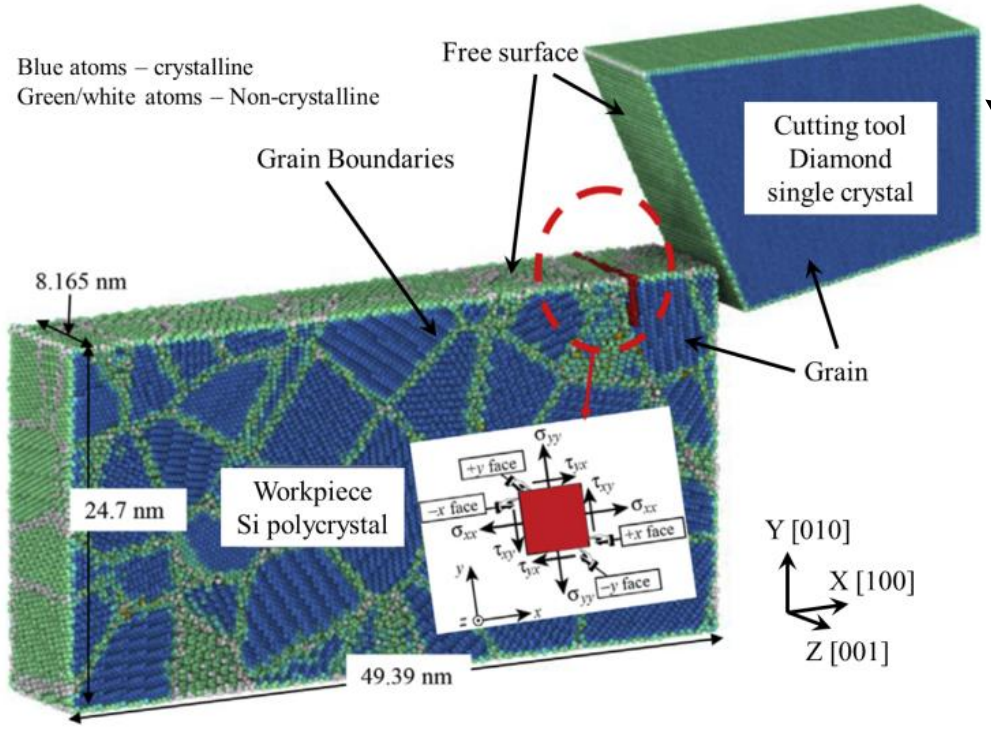
Contours of hydrostatic pressure at 20 % reduction in height with f = 0.05 and R/H = 1/2



Contours of normal stress at 20 % reduction in height with f = 0.05 and R/H = 1/2

Molecular dynamics

- Case: Influence of Microstructure on Cutting Behavior of Silicon (Goel 2016)
- Nanometric cutting of single crystalline (SC) and polycrystalline (PC) silicon
- MD performed using large-scale atomic/molecular massively parallel simulator (LAMMPS)



MD simulation model of nanometric cutting of polycrystalline silicon.

Rigid boundary condition for tool

Sharp cutting edge used

Si atom bond: covalent nature

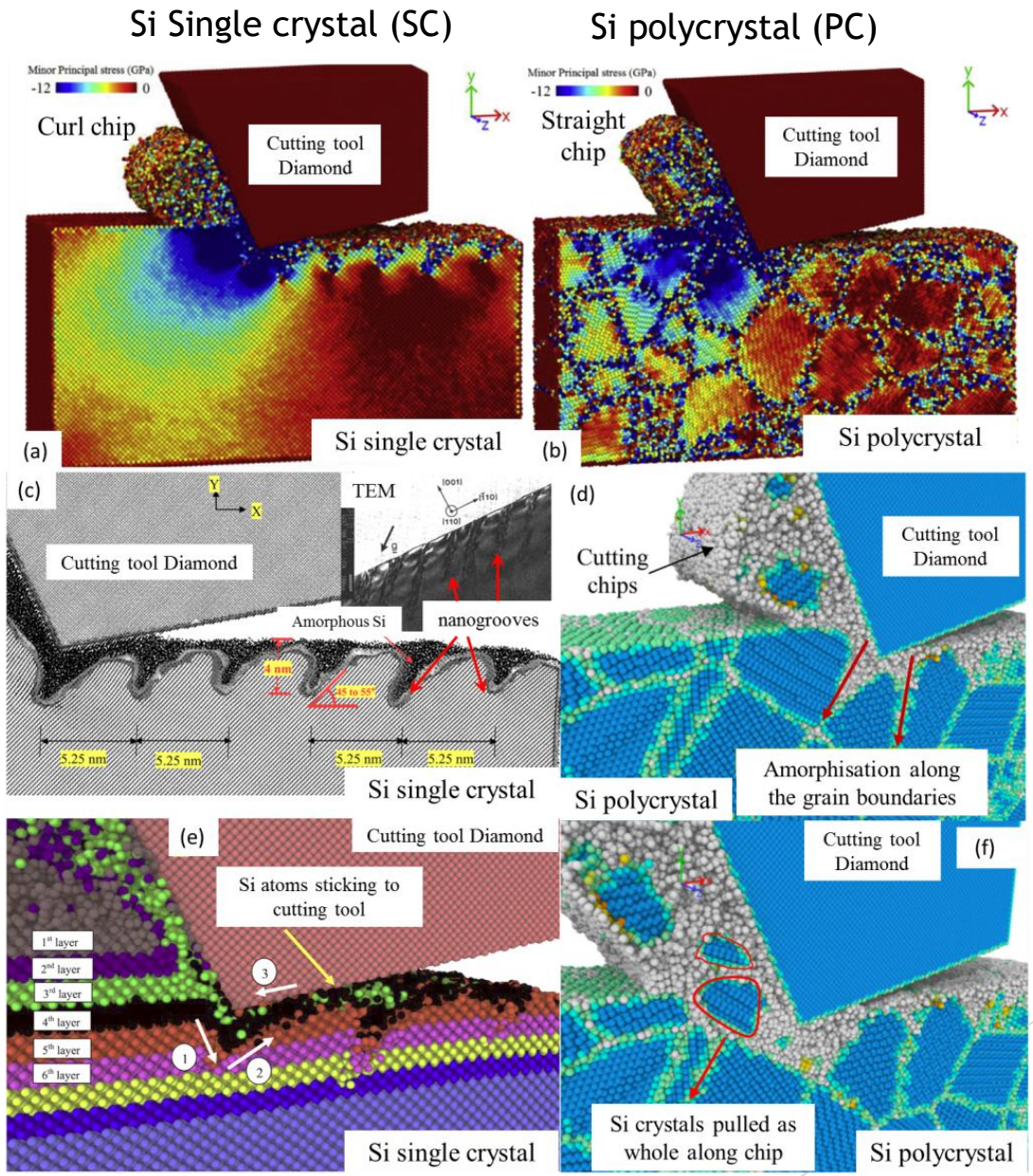
Analytical bond-order potential (ABOP) used (Erhart 2005)

MD: Results

Variation in the minor principal stress (compressive) during cutting

Periodic nano-grooves formation (45° - 55° to the cutting direction) in cutting zone like that observed in experiments

Vorticity path observed from atomic displacements



Cutting force: SC>PC

Cutting curl: SC more curled than in PC

Amorphization and grain pull out observed in simulation of PC Si

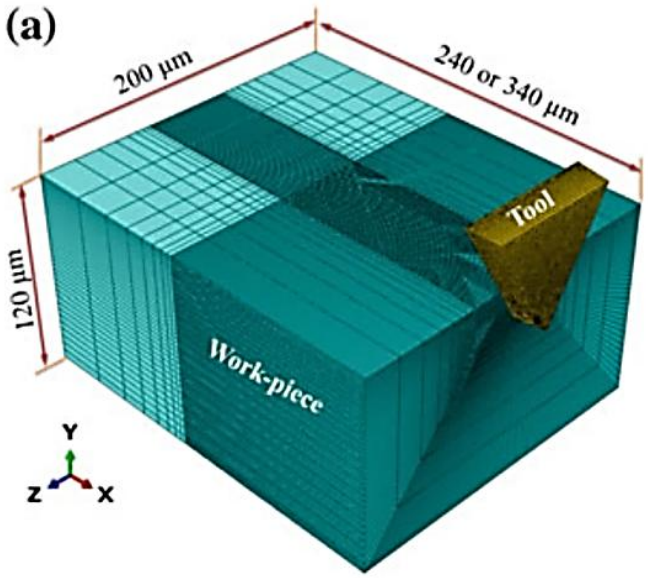
In PC, plastic deformation concentrated along grain boundaries

CP FEM

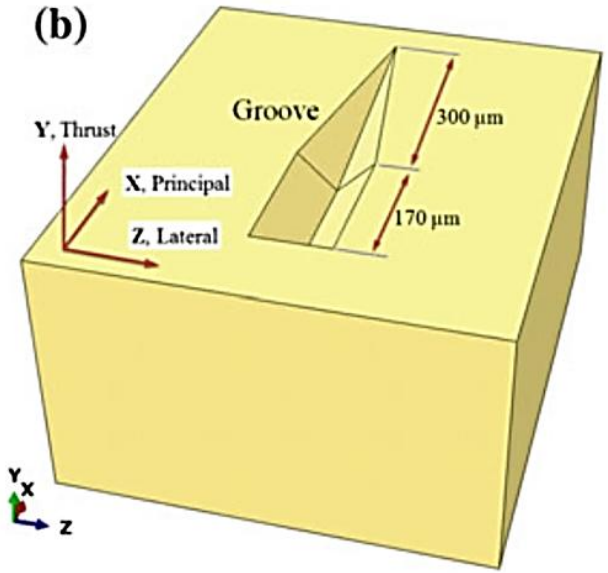
- Case: Crystal-plasticity Simulation of Micromachining of Single-crystal Metal (Liu 2016)
- Micro-scratching of single-crystal copper (FCC structure)
- Two-stage micro-scratching: a linearly increasing cutting depth upto 18 μm (engagement stage) followed by a constant cutting depth of 18 μm (full-engagement stage)

Element: C3D8R

Finer mesh near cutting zone



FE model



Two-step micro-scratching

Element deletion used for material removal simulation

Cutting tool:
60° wedge angle
6.25° clearance angle

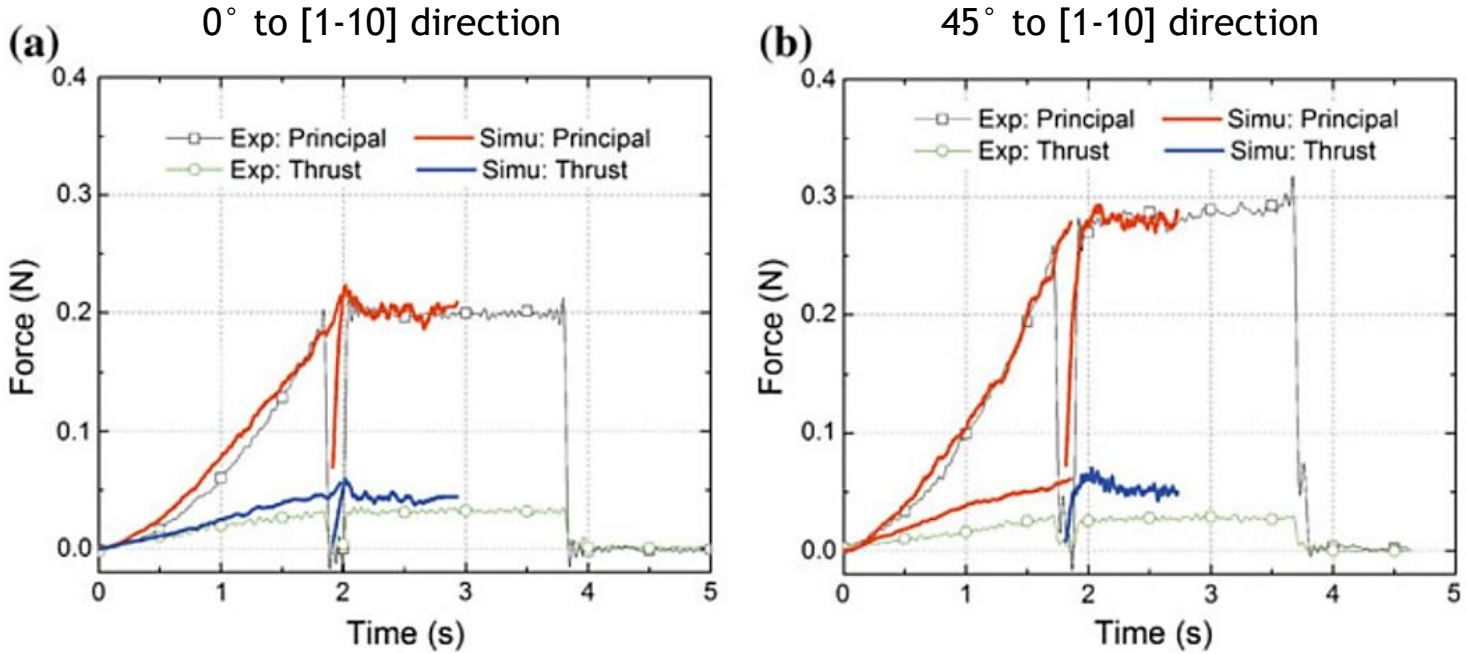
Cutting velocity: 10mm/min

Cu single crystal:
(110) Plane parallel to cutting plane

Cutting direction:
0° to [1-10] direction (symmetric)
45° to [1-10] direction (asymmetric)

Cp FEM: REsults

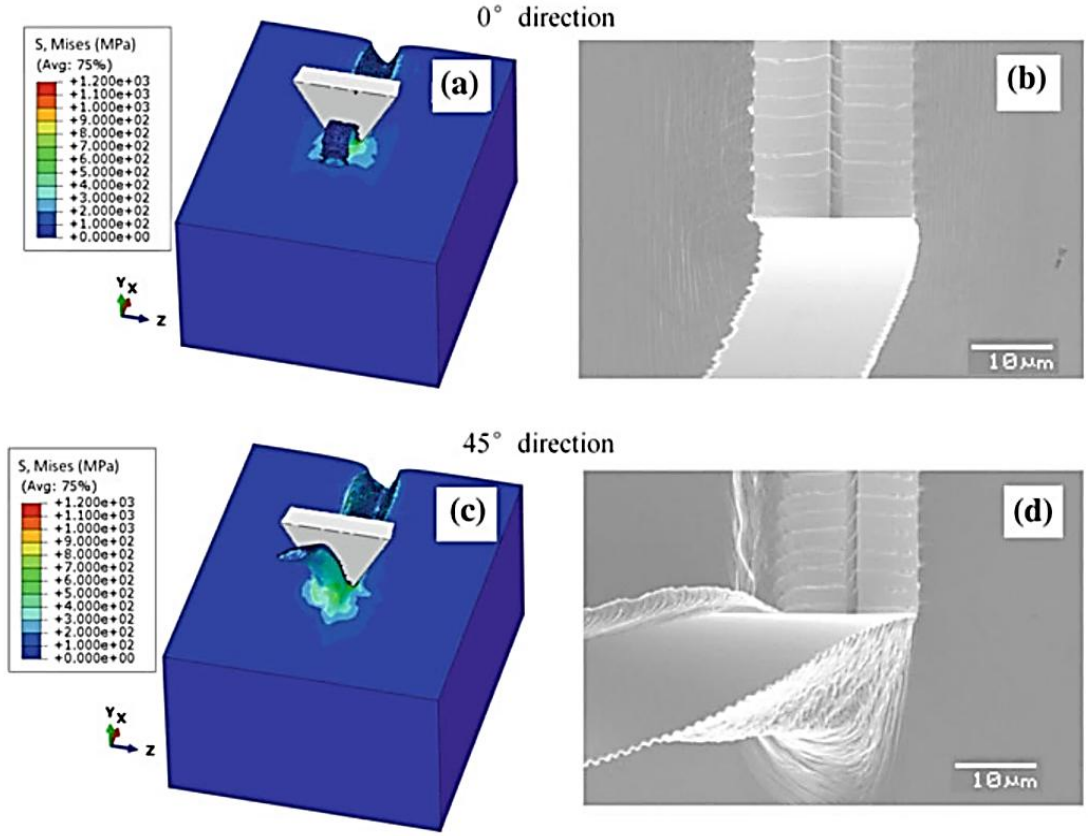
Cutting Force



- Principal force matches well with the experimental results.
- Principal force for the 45° direction was 40% higher than that for 0° direction and shows the anisotropy in the material.
- Thrust forces prediction are relatively poor and not dependent on cutting direction

CP FEM: Results

Chip morphology



Directional dependence of chip morphology is well captured in simulation

For 0° direction, chip is symmetric

For 45° direction, chip is asymmetric

Chip morphology is governed by symmetry along cutting direction

- Crystal possess a two-fold symmetry along [1-10] direction (0° direction), hence, activated slip systems were symmetrical with respect to the cutting direction 0° leading to symmetrical chip formation
- no crystal symmetry exists along the cutting 45° direction, which explains the asymmetrical chip

MD vs CP FEM

MD:

- ✓ Useful when underlying deformation mechanism not well understood
- ✓ Gives fundamental insight of the deformation processes
- ✓ No prior knowledge of system evolution required

- ○ Increased complexity of the potential and computational cost for multi-component system
- ○ Timescale (~ns) and domain size (~100 nm) are very limited
- ○ Unrealistic strain rate (10^8 s^{-1} and above) far from real-world situation simulated

Courtesy: Prof. Anish Roy, UK

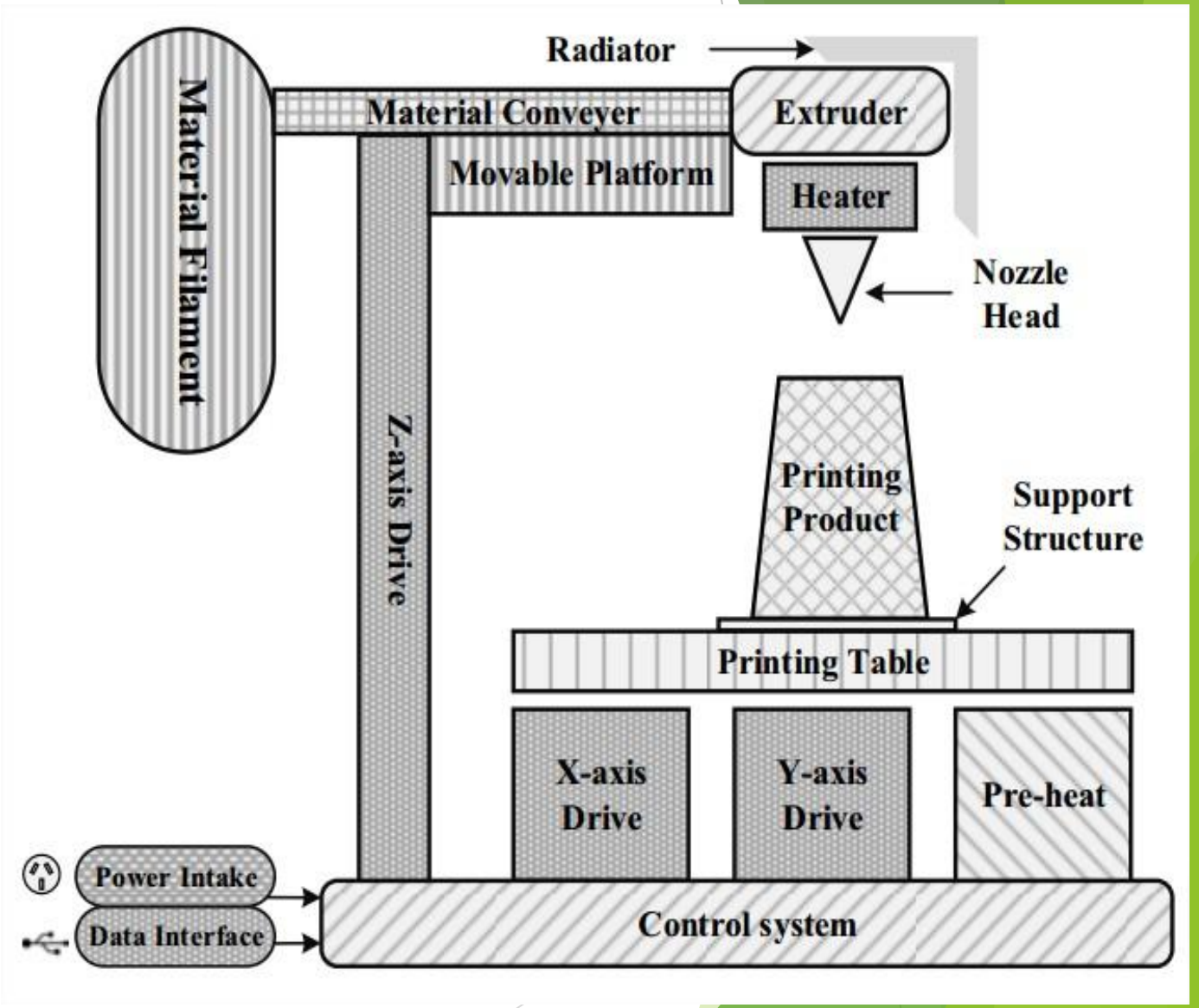
CP FEM:

- ✓ Computationally more efficient
- ✓ Capability to model reasonable spatial and time domains
- ✓ Various deformation modes (slip, twin, etc) can be modelled
- ✓ Material with different crystal structure (FCC, HCP, BCC) can be modelled
- ✓ Deformation mode dependency on various parameters (eg. Size, temperature, strain rate, etc) can be incorporated
- ✓ Capability to investigate individual deformation mode (separately)

- ○ Phenomenological material laws need to be established for new material via initial experimental investigation
- ○ Involves identification of deformation modes and calibration of material model

Additive Manufacturing (3D printing):
A disruptive technology

- 3D printing (3DP) or Additive Manufacturing produces three dimensional physical objects through layer-by-layer formation based on a digital blueprint (CAD file).
- 3DP has proved to be compatible with industrial manufacturing beyond prototyping.



Source: Tao Peng, Analysis of energy utilization in a 3D printing process, Procedia CIRP 40 (2016) 62 – 67

Figure A schematic of a 3DP

Classification of AM

Based on raw material			Based on layer formation
Solid based	**Powder based**	**Liquid based**	
Fused deposition modelling (FDM)	3D printing (3DP)	Stereolithography (SLA)	Extrusion (FDM)
Laminated object manufacturing (LOM)	Selective laser sintering (SLS)	Polyjet	Sheet lamination (LOM)
	Selective laser melting (SLM)		Beam deposition (LENS)
	Electron beam melting (EBM)		Powder bed fusion (SLS, SLM)
			Binder jet printing (3DP)
			Photopolymerisation (SLA)
			Direct write technology

Stereolithography (SLA)

- Patented by Charles W. Hull in the year 1986 (Hull 1986).
- Relies on the use of photopolymers and curable resins.
- The laser traces the cross section of the pattern of the product.
- A build platform (substrate) is attached to support structures.
- Platform is lowered for coating.
- Substrate is again raised where the blade wipes the resin and a layer is left on the part.
- For ensuring proper settlement of the liquid, the build platform is lowered by a layer (Renap and Kruth 2010).

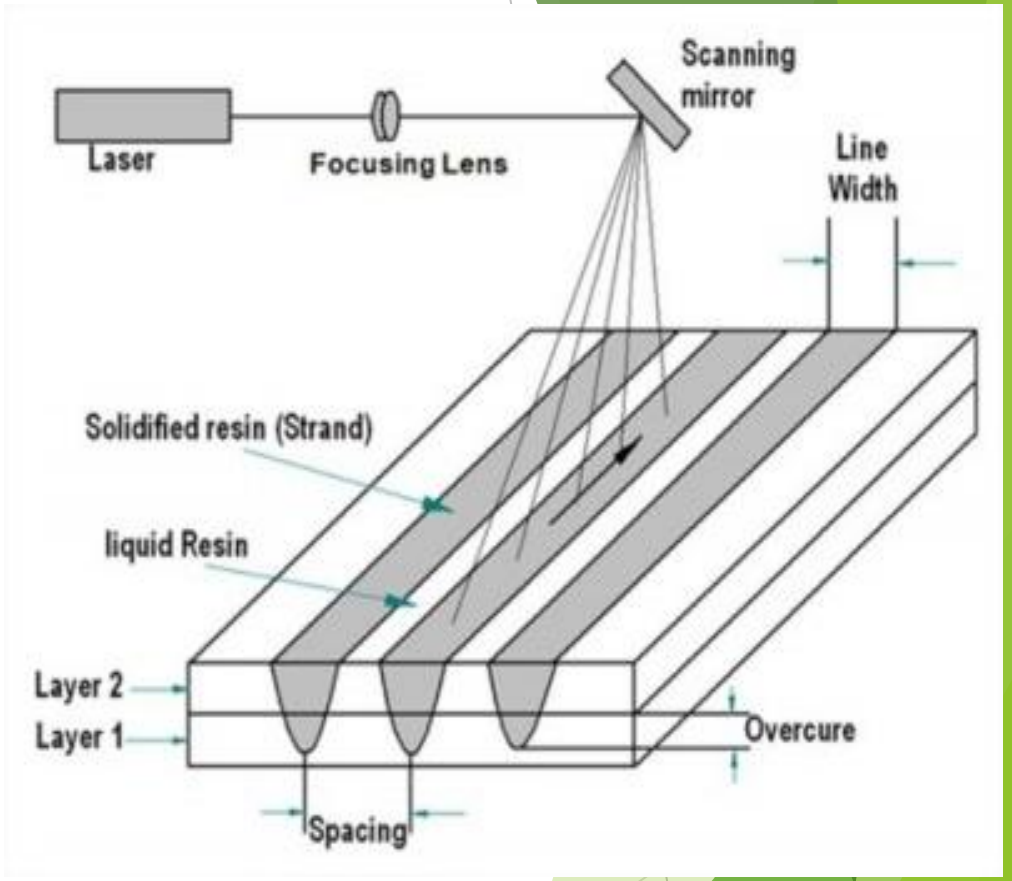


Figure A schematic of SLA (Raju et al. 2014)

Source: Raju, B.S., Shekar, U.C., Venkateswarlu, K. and Drakashayani, D.N., 2014. Establishment of Process model for rapid prototyping technique (Stereolithography) to enhance the part quality by Taguchi method. *Procedia Technology*, 14, pp.380-389.

Selective laser sintering (SLS)

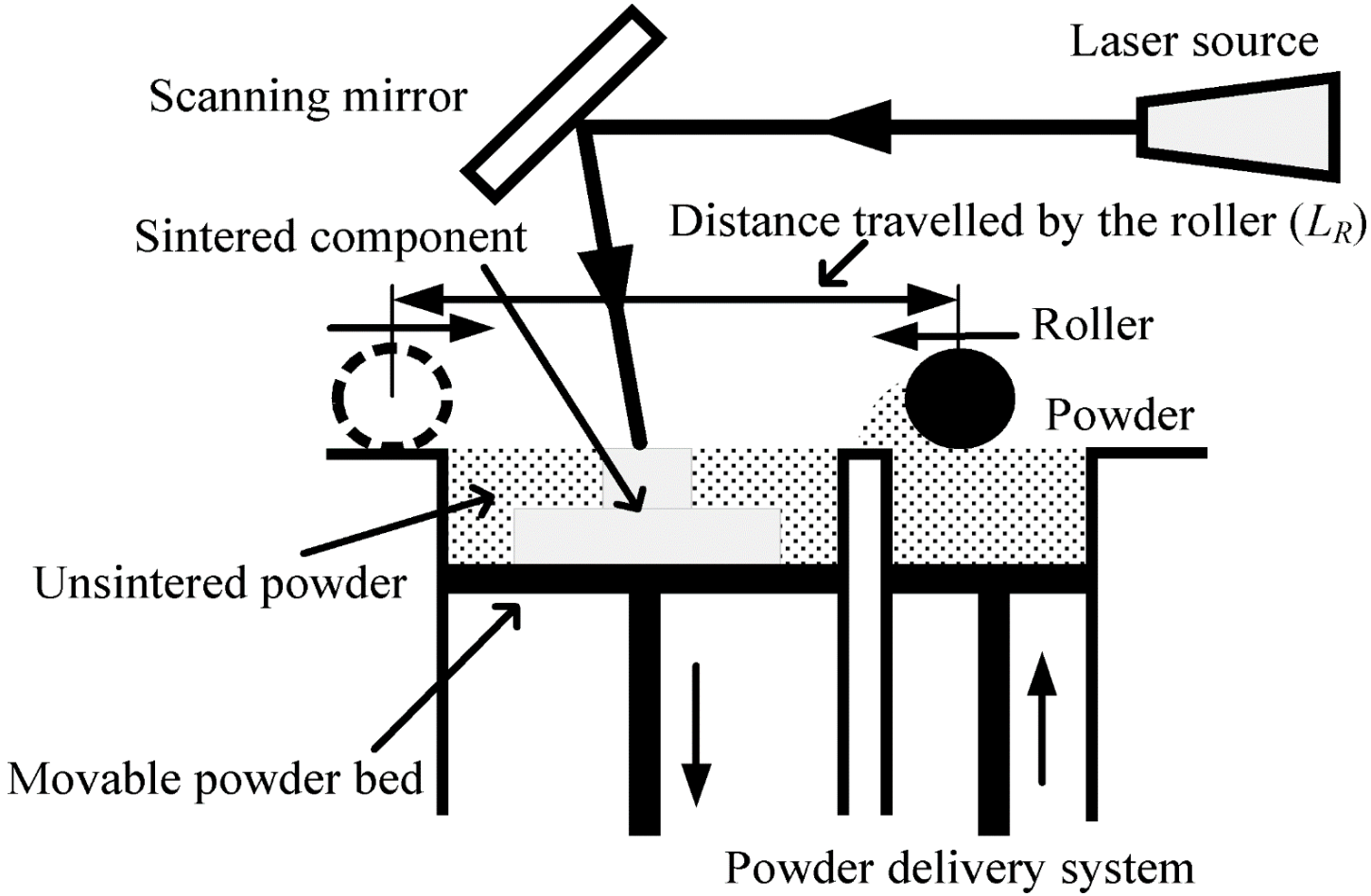


Figure A schematic of powder based fusion technology with vacuum chamber

- Patented by Carl R. Deckard in the year 1989 High power laser beam fuses the powders at a molecular level in a particular area of the build material.

Selective laser sintering (SLS)

- Powder bed is heated nearly to the melting point of the material to allow proper sintering of the powders (Klocke et al. 1995).
- Rollers present on each side of the build platform provide the direction to the powders for each subsequent layer.
- The particles lying in the powder bed controlled by a movable piston is free to move in the z direction.
- The movement takes place according to the prescribed layer thickness.
- Selective laser melting (SLM) employs similar principle.

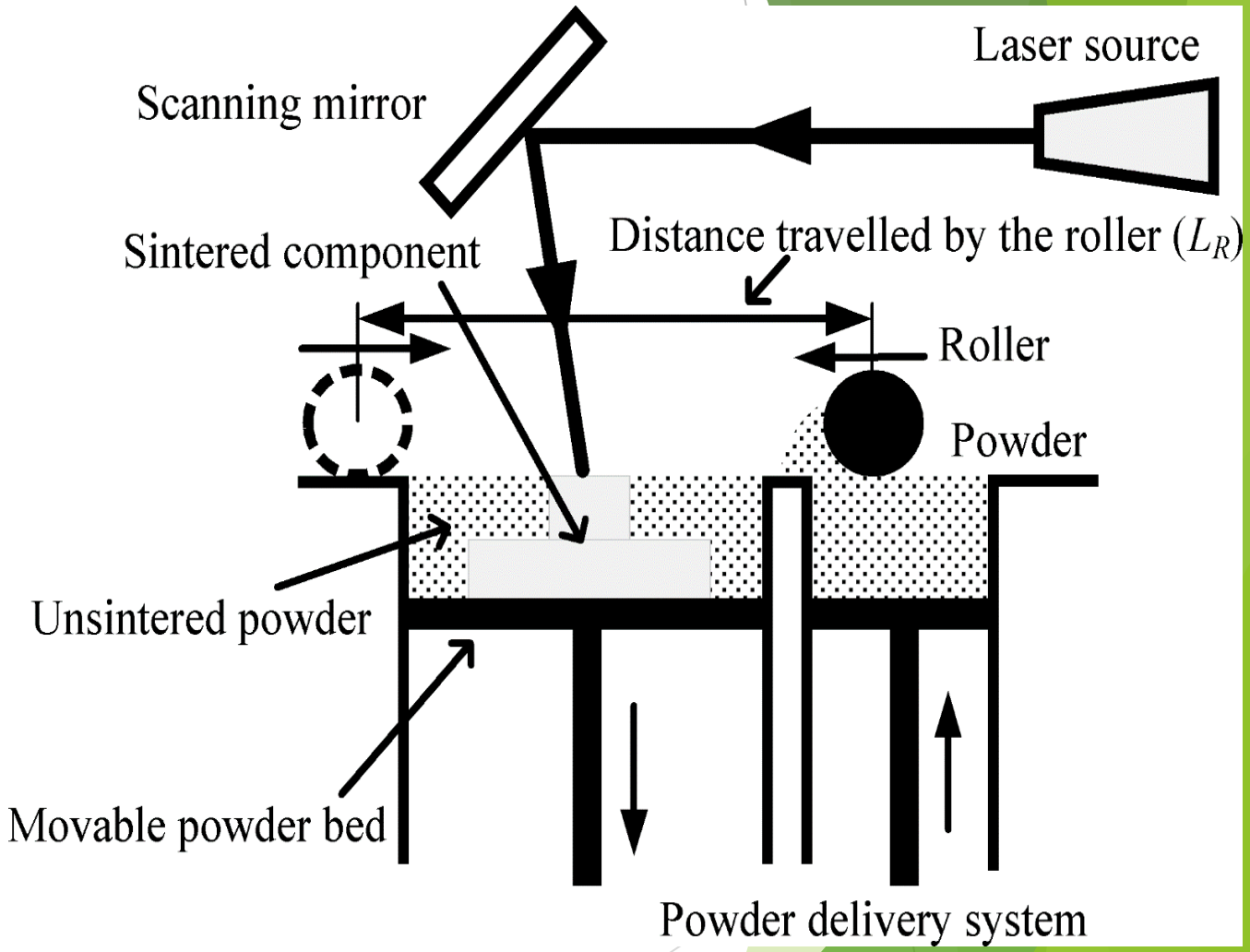


Figure A schematic of powder based fusion technology with vacuum chamber

3D printing (3DP)

- Powdered material are deposited on the substrate and fused by a binder sprayed by the nozzle.
- Powders are misted with water droplets by capillary action (Sachs et al. 1993).
- The size, shape and rheology of the powder along with the interaction with the binder play a dominating role.
- The built part is subjected to firing for sintering to take place and strengthen the bonding.

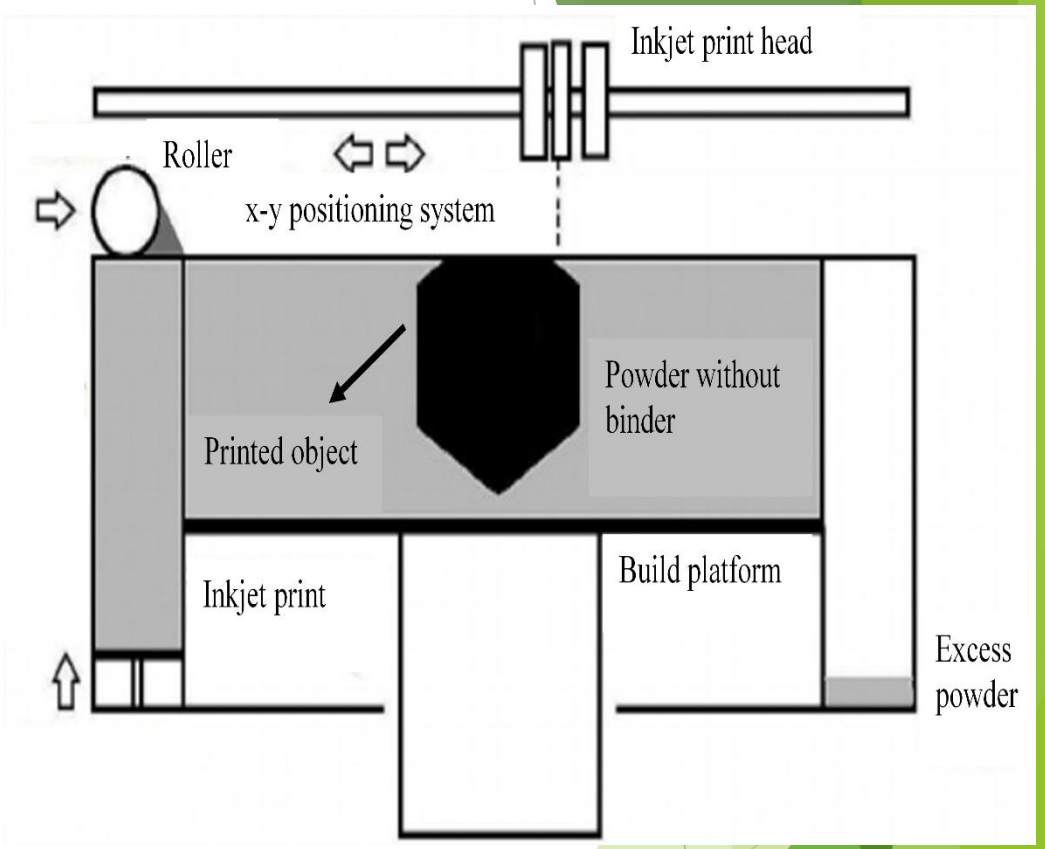


Figure A schematic of binder jetting 3D printing process (Holland et al. 2018)

Source: Holland, S., Foster, T., MacNaughtan, W. and Tuck, C., 2018. Design and characterisation of food grade powders and inks for microstructure control using 3D printing. *Journal of Food Engineering*, 220, pp.12-19.

Laminated object manufacturing (LOM)

- Patented by Michael Feygin in the year 1988.
- Continuous supply of raw material sheet from a feed roll gets bonded to the previous layers.
- A heated roller is passed over the layers where a heat sensitive adhesive which is already applied to the surface of the sheet.
- Once the layer is built, the built platform is lowered where the excess (waste) material gets rolled by a second roller to allow fresh material come over the build platform.

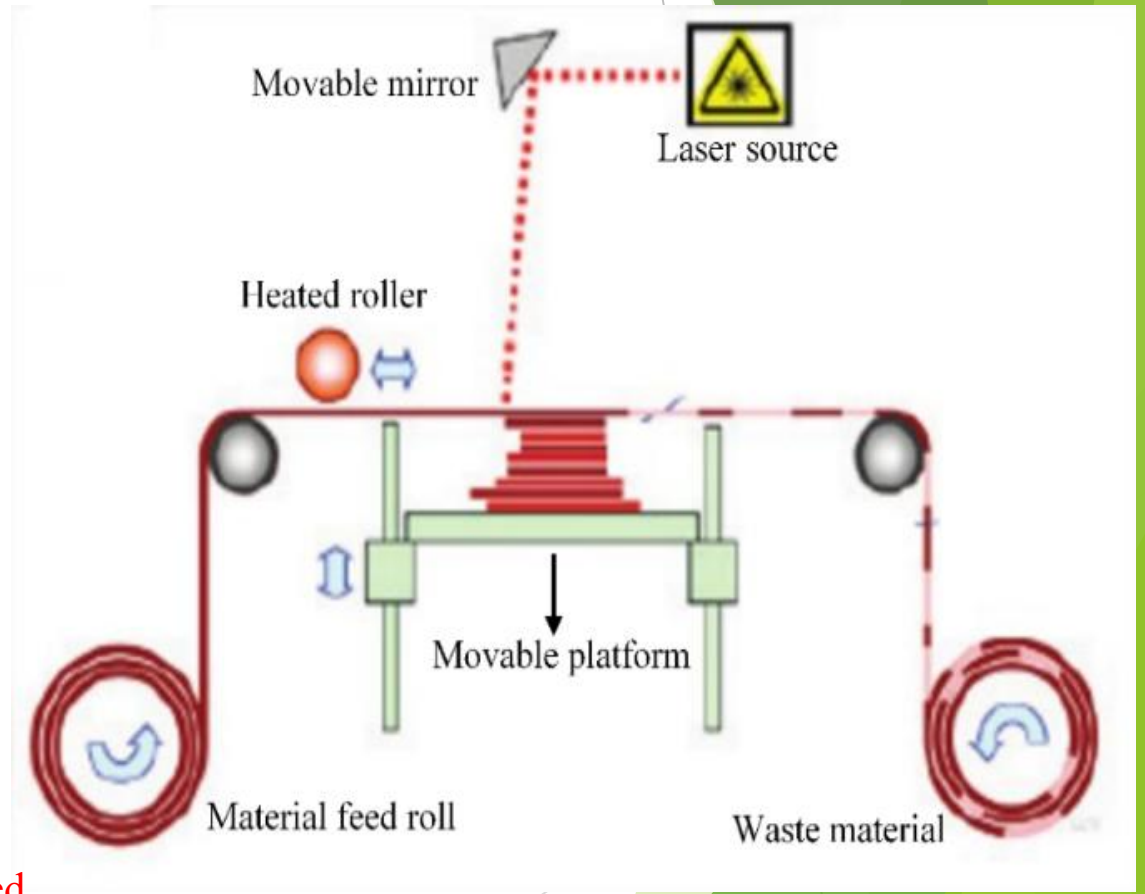


Figure A schematic of LOM (Zhang and Wang 2017)

Source: Zhang, Y. and Wang, J., 2017. Fabrication of Functionally Graded Porous Polymer Structures using Thermal Bonding Lamination Techniques. *Procedia Manufacturing*, 10, pp.866-875.

Laser engineered net shaping (LENS)

- Uses an extremely powerful source of either laser or electron beam.
- The nozzle is mounted on a multi axis (3-5) arm that can move in multiple direction.
- The melted metal powders are infused in a substrate to create a layer.
- For solidification to occur and create a layer, the substrate is moved underneath the laser beam (Gibson et al. 2009).
- The uniqueness of this process lies in its ability to use multiple materials at the same time.

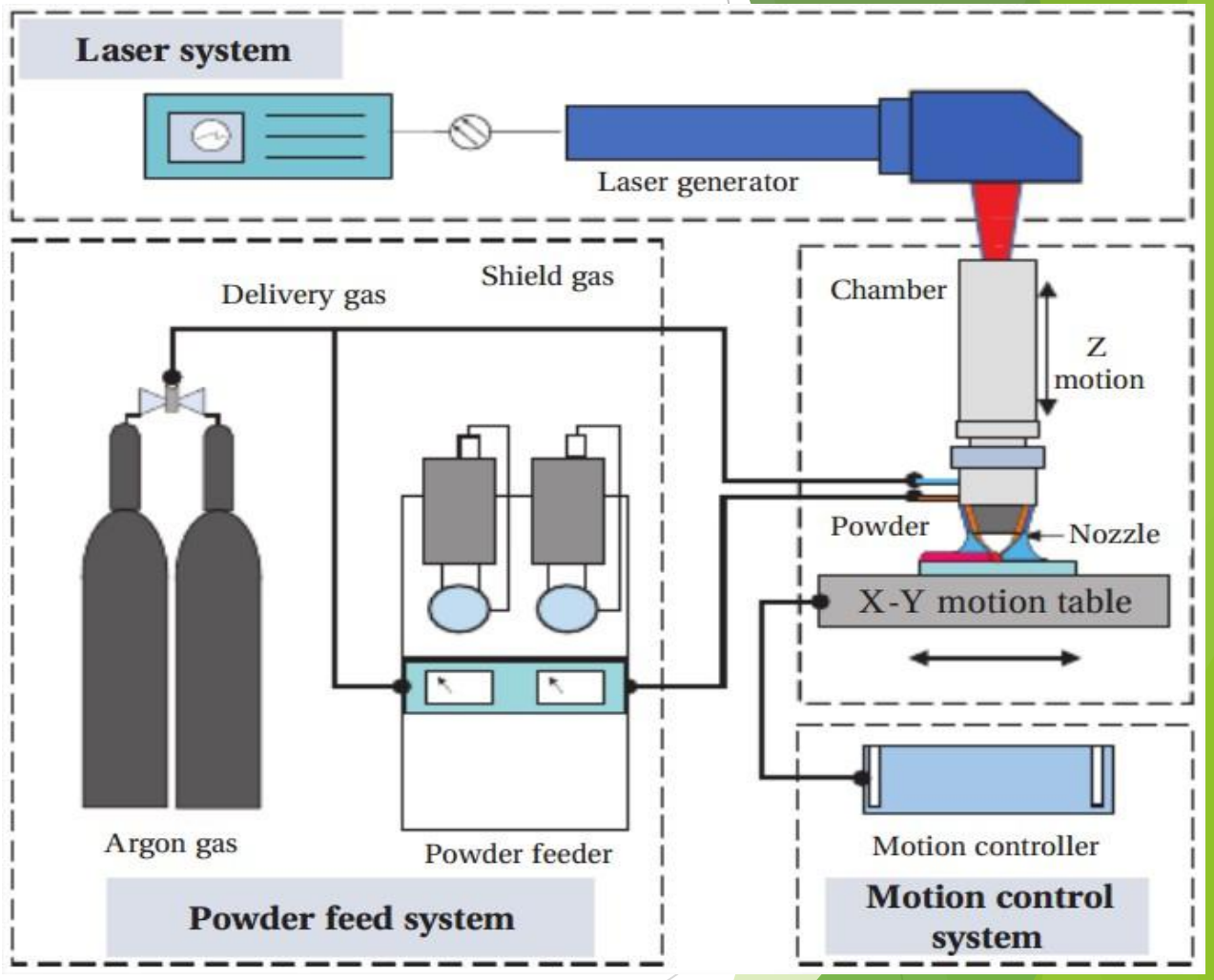


Figure A schematic of LENS (Liu et al. 2017)

Source: Liu, Z., Cong, W., Kim, H., Ning, F., Jiang, Q., Li, T., Zhang, H.C. and Zhou, Y., 2017. Feasibility Exploration of Superalloys for AISI 4140 Steel Repairing using Laser Engineered Net Shaping. *Procedia Manufacturing*, 10, pp.912-922.

Features of different AM processes (Pham and Gault 1998; Huang et al. 2013; Huang et al. 2015)

Category	Process	Raw material	Layer thickness (μm)	Laser requirement (Yes/No)
Material extrusion	FDM	Polycarbonate (PC), ABS, PLA	100–250	No
Sheet lamination	LOM	PVC sheet, aluminium sheet	80–200	Yes
Binder jetting	3D printing	Polyvinyl alcohol, metal oxide powder, polymeric powder	100–400	No
Material jetting	Inkjet printing	Acrylic powder, calcium carbonate	5–200	No

Category	Process	Raw material	Layer thickness (μm)	Laser requirement (Yes/No)
Powder bed fusion	SLS	Titanium alloys, ceramics powders, polyamide	25–92	Yes
	SLM	Stainless steel, titanium, aluminium	75–150	Yes
Direct energy deposition	Laser engineered net shaping (LENS)	Functionally graded materials, stainless steel	200–500	Yes
Photopolymerisation	SLA	Plastics, glass, ceramics	50–100	Yes

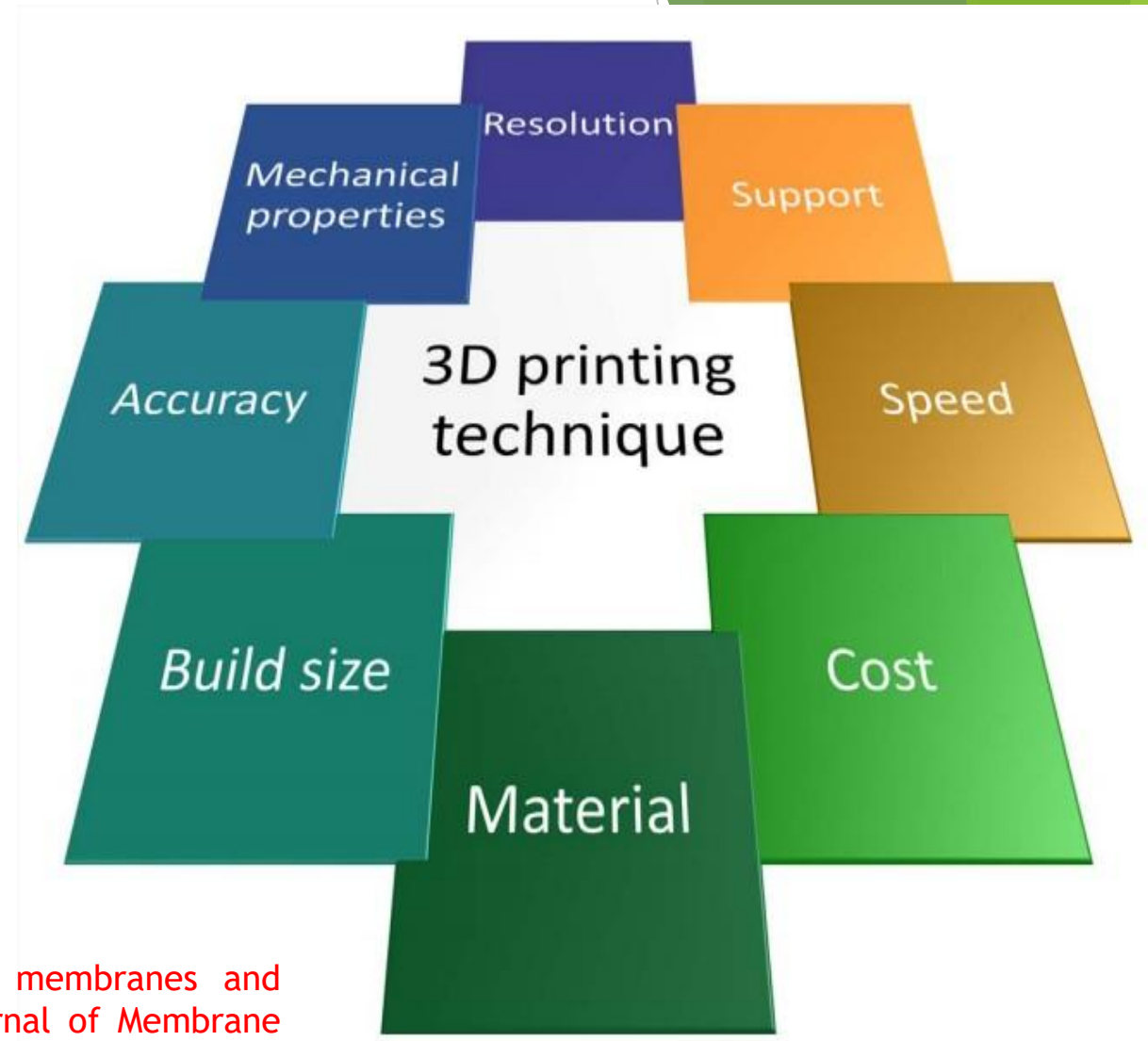
Application and comparison of various AM processes

(Pham and Gault 1998; Ngo et al. 2018; Prakash et al. 2018)

Category	Applications	Benefits	Limitations
SLA	Fabricate nanocomposites Biomedical industry Hybrid polymer ceramics	Good surface finish Fine resolution Thin layer thickness	Limited range of materials Small product size Expensive
FDM	Fabricate prostheses Advanced composite parts Rapid prototyping	High speed Low cost Sustainable material	Layer by layer appearance Temperature fluctuation Inability to get sharp edges
SLS	Light weight structures Electronic components Defence hardware	Building complex parts Recycling of powders Better functionality	Poor surface finish Relatively slow process Material changeover issue

Category	Applications	Benefits	Limitations
LENS	Retrofitting Aerospace and automobiles Cladding	Less manufacturing time High strength Quality microstructure	Dense support structure Limited use of materials Less accuracy
3DP	Moulds for metal casting Ceramic preform in composites Structural parts	Fast fabrication Low material cost	Fragile and porous parts Rough surface finish
LOM	Investment casting pattern Metal filled tapes Concept verification	Inexpensive Produce large scale model No toxic fumes	Less dimensional stability Generate internal cavities Lengthy post processing

- Digitalization of manufacturing by 3D printing is an important aspect of Third Industrial Revolution.
- Despite prospective benefits of 3D printing, research on sustainability of 3D printing needs to be performed before the market explodes.



Source: Low et al., Perspective on 3D printing of separation membranes and comparison to related unconventional fabrication techniques, Journal of Membrane Science, 523 (2017) 596–613

Figure Properties of 3DP techniques

Additive manufacturing

- How can it be competitive with other manufacturing processes?
- Speed
- Mechanical Properties and their characterization
- Cost
- Availability of raw materials
- Overall sustainability

Industry 5.0

- Industry 5.0 is a proposed fifth industrial revolution that builds upon Industry 4.0, emphasizing human-centricity and sustainability. It aims to create a more sustainable and resilient industry by balancing technological advancements with the well-being of workers and the planet.

Advantages

- **Human-Centric Approach**: Focuses on collaboration between humans and machines (especially robots), enhancing worker creativity, satisfaction, and safety.
- **Personalization**: Enables mass customization of products to meet individual customer needs, combining efficiency with tailored experiences.
- **Sustainability**: Emphasizes eco-friendly technologies and circular economy practices,
- aiming to reduce waste and environmental impact.
- **Resilience**: Builds systems that can adapt to disruptions (like pandemics or supply chain shocks) through smart automation and human flexibility.
- **Enhanced Innovation**: Encourages co-creation between humans and AI/robots, driving rapid and diverse innovation across sectors.
- **Better Work Conditions**: Automates repetitive and dangerous tasks, improving job quality and worker well-being.
- **Social Responsibility**: Aligns technological progress with societal goals, including inclusion, ethical AI, and digital sovereignty.

Aspect	Industry 4.0	Industry 5.0
Core Focus	Automation, efficiency, and productivity	Human-centric, sustainable, and resilient manufacturing
Role of Humans	Minimized; replaced by machines and automation	Central; collaborate with machines (cobots)
Production Type	Smart mass production	Smart **mass customization**
Technology Emphasis	IoT, AI, big data, robotics, cyber-physical systems	Human-machine collaboration, ethical AI, emotional intelligence
Sustainability	Limited focus	Strong emphasis on green and circular economy
Resilience	Vulnerable to disruptions	Designed to adapt and recover quickly
Innovation Approach	Machine-led innovation	Co-creation between humans and machines
Ethical Consideration	Rarely addressed	Strong emphasis on ethics, inclusion, and social impact
End Goal	Smart factories	Harmonious coexistence of technology, people, and the planet